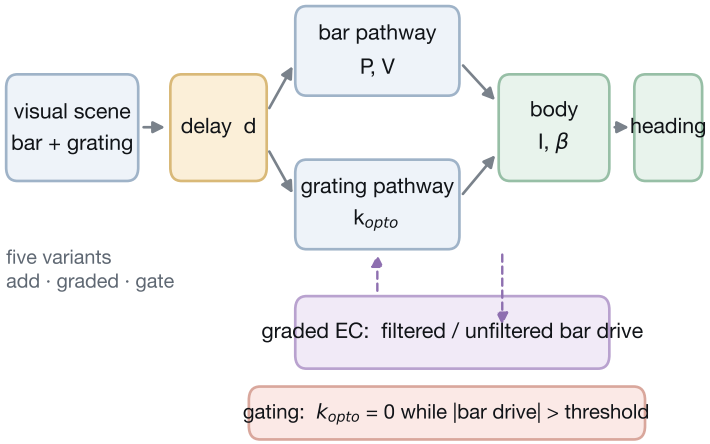
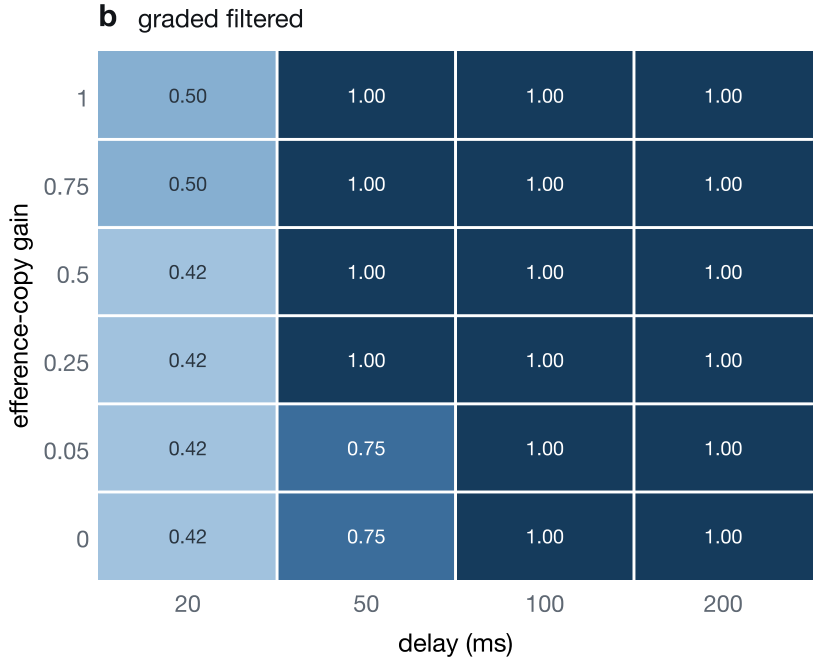
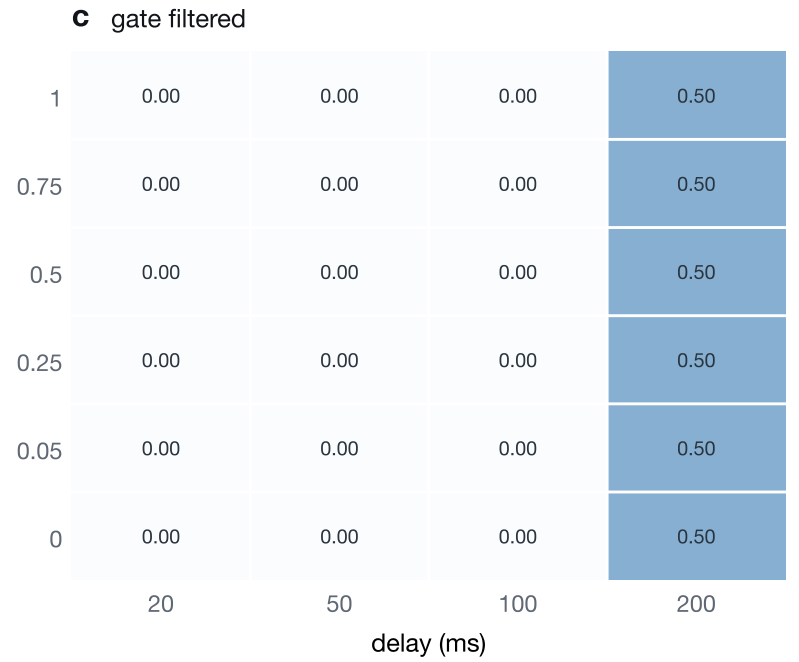
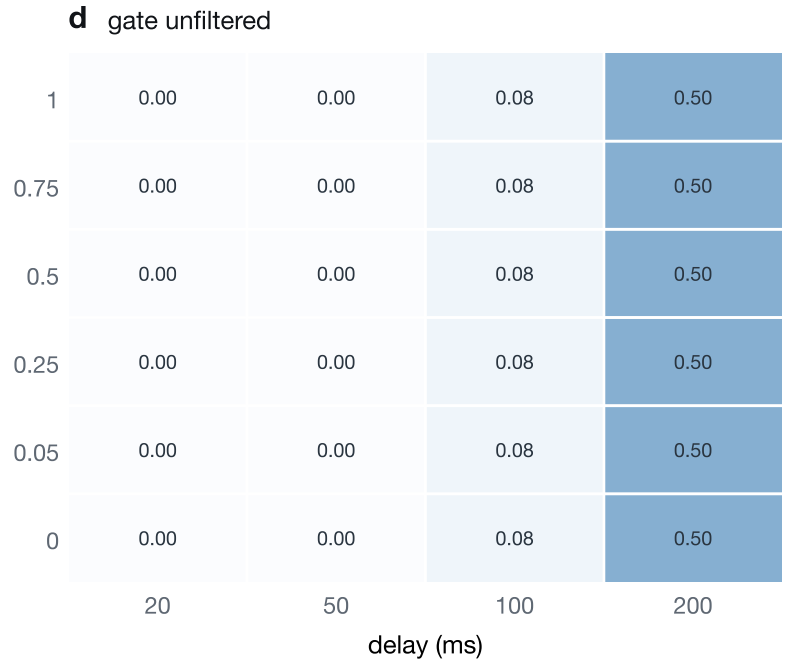
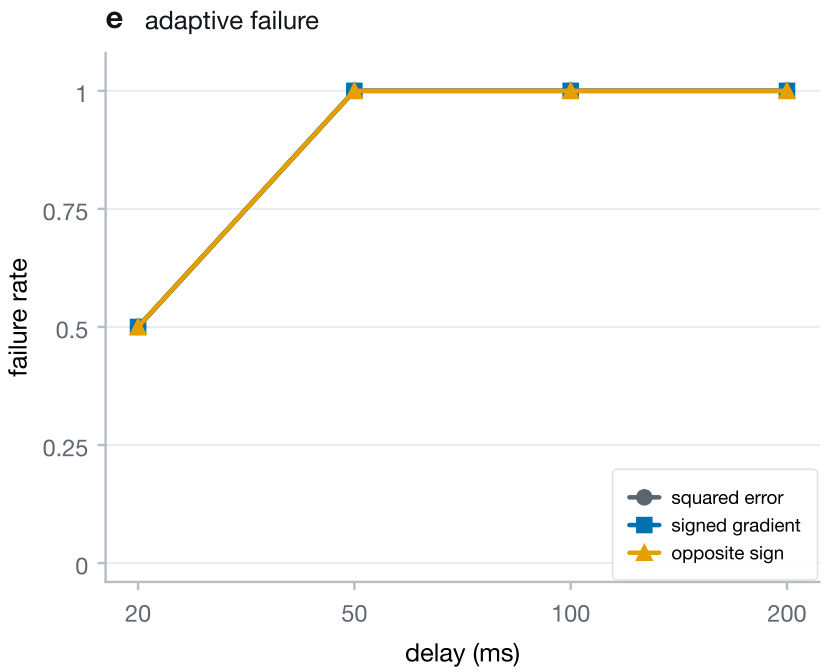
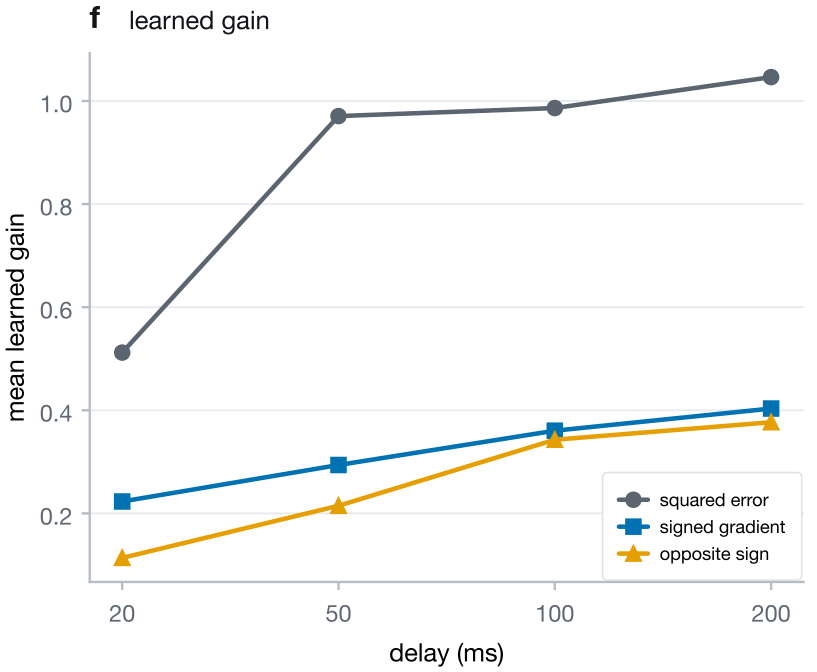

visual scene
bar + grating
delay d
bar pathway
P, V
grating pathway
$k_{opto}$
body
I, β
heading
five variants
add · graded · gate
graded EC: filtered / unfiltered bar drive
gating: $k_{opto}$ = 0 while |bar drive| > threshold


b graded filtered
efference-copy gain
1
0.75
0.5
0.25
0.05
0
0.50 1.00 1.00 1.00
0.50 1.00 1.00 1.00
0.42 1.00 1.00 1.00
0.42 1.00 1.00 1.00
0.42 0.75 1.00 1.00
0.42 0.75 1.00 1.00
20 50 100 200
delay (ms)


c gate filtered
1
0.75
0.5
0.25
0.05
0
0.00 0.00 0.00 0.50
20 50 100 200
delay (ms)


d gate unfiltered
1
0.75
0.5
0.25
0.05
0
0.00 0.00 0.08 0.50
20 50 100 200
delay (ms)


e adaptive failure
failure rate
1
0.75
0.5
0.25
0
20 50 100 200
delay (ms)
squared error
signed gradient
opposite sign


f learned gain
mean learned gain
1.0
0.8
0.6
0.4
0.2
20 50 100 200
delay (ms)
squared error
signed gradient
opposite sign